Supporting Information

Appendices S1 and S2 derive the mathematical and statistical foundations for our approach respectively. Appendix S3 evaluates the performance of our approach. Appendix S4 describes the data and data analysis.

Appendix S1: Derivation of the coevolutionary model

S1.1: Selection

We assume the biotic effects on fitness act on the fecundity of an individual while the abiotic effects act on the viability of an individual with $W_{B,i}$ as the expected reproductive output of an individual of species i given that it survives to reproduce and $W_{A,i}$ being the probability of surviving until reproduction for an individual of species i. Specifically, assuming that abiotic selection is Gaussian with a constant phenotypic optimum θ_i, tolerance α_i and maximum $0 < p_i \leq 1$, we define the probability of a member of species i with trait z_i surviving abiotic selection by

$$W_{A,i}(z_i) \equiv p_i \exp\left(-\frac{(\theta_i - z_i)^2}{2\alpha_i}\right). \tag{1}$$

To model fecundity selection based on the interactions between species we assume the interactions are mediated by what we call an offset matching mechanism. In this model biotic fitness is maximized for an individual when its trait is offset from the trait of the individual it is interacting with by some constant δ. A simple example of an optimal offset comes from considering the interaction between long-tubed flowers and the long-proboscid flies that visit them. The biotic component of fitness for the fly is maximized when its proboscis is slightly longer than the nectar tube depth of the flower, allowing the fly to easily extract its nectar reward. The difference between tube depth and proboscis length that maximizes the flies biotic fitness component is the optimal offset for the fly. Note how this differs from a "bigger is better" situation since fitness with an optimal offset is unimodal and therefore does not increase indefinitely with larger (or lesser) trait values.

To capture this we assume biotic selection is also Gaussian but with γ_i as the tolerance and $z_j + \delta_i$ as the phenotypic optimum where z_j is the trait value of the individual being encountered and δ is the optimal offset in trait value that maximizes the biotic component of fitness. By assuming there is an upper bound on the potential number of offspring (which we will denote by c_i), we define the proportion of this upper bound achieved by an individual

of species i given that it has encountered an individual of species j by

$$s_i(z_i, z_j) \equiv \exp\left(-\frac{(z_j + \delta - z_i)^2}{2\gamma_i}\right). \tag{2}$$

Next, we assume individuals encounter each other at random (with respect to their trait values) and that traits are normally distributed with densities denoted by $\phi_1(z_1)$ and $\phi_2(z_2)$. We denote the means of these distributions by $\bar{z}_1, \bar{z}_2$ and variances by σ_1^2, σ_2^2. Then for an individual in species i with trait z_i, the expected proportion of the maximal reproductive output is defined as

$$\begin{aligned} b_i(z_i) &\equiv \int_{\mathbb{R}} \phi_j(z_i) s(z_i, z_j) dz_j \\ &= \int_{\mathbb{R}} \frac{1}{\sqrt{2\pi\sigma_j^2}} \exp\left(-\frac{(\bar{z}_j - z_j)^2}{2\sigma_j^2}\right) \exp\left(-\frac{(z_j + \delta - z_i)^2}{2\gamma_i}\right) dz_j \\ &= \sqrt{\frac{\gamma_i}{\gamma_i + \sigma_j^2}} \exp\left(-\frac{(\bar{z}_j + \delta - z_i)^2}{2(\gamma_i + \sigma_j^2)}\right). \end{aligned} \tag{3}$$

Proving this statement is a matter of elucidating the algebra of Gaussian functions. For example, if $f_1(x)$ and $f_2(x)$ are Gaussian functions of x then $f_1(x)f_2(x)$ will also be a Gaussian function of x. Similarly, if $f_1(x)$ and $f_2(x)$ are the densities of two normal distributions then, $f_1(x)f_2(x)$ will be proportional to a density of another normal distribution. Specifically, we have

$$\begin{aligned} &\frac{1}{\sqrt{2\pi\sigma_1^2}} \exp\left(-\frac{(\mu_1 - x)^2}{2\sigma_1^2}\right) \frac{1}{\sqrt{2\pi\sigma_2^2}} \exp\left(-\frac{(\mu_2 - x)^2}{2\sigma_2^2}\right) \\ &= \frac{1}{\sqrt{2\pi(\sigma_1^2 + \sigma_2^2)}} \exp\left(-\frac{(\mu_1 - \mu_2)^2}{2(\sigma_1^2 + \sigma_2^2)}\right) \frac{1}{\sqrt{2\pi\tilde{\sigma}^2}} \exp\left(-\frac{(\tilde{\mu} - x)^2}{2\tilde{\sigma}^2}\right) \end{aligned} \tag{4}$$

where

$$\tilde{\mu} = \frac{\sigma_2^2\mu_1 + \sigma_1^2\mu_2}{\sigma_2^2 + \sigma_1^2}, \ \tilde{\sigma}^2 = \frac{\sigma_1^2\sigma_2^2}{\sigma_1^2 + \sigma_2^2}.$$

To demonstrate this equality rigorously requires a lengthy calculation involving the Fourier transform and can be found elsewhere. We therefore omit it here. This result can be used to evaluate equation (3) and will be employed a great deal throughout the rest of the derivation.

Putting the pieces together, biotic fitness for an individual of species i with trait z_i can then be written as

$$W_{B,i}(z_i) = c_i b_i(z_i) = c_i \sqrt{\frac{\gamma_i}{\gamma_i + \sigma_j^2}} \exp\left(-\frac{(\bar{z}_j + \delta - z_i)^2}{2(\gamma_i + \sigma_j^2)}\right). \tag{5}$$

Hence the expected reproductive output of an individual with trait z_i in species i is expressed as

$$W_i(z_i) = W_{B,i}(z_i) W_{A,i}(z_i) = c_i p_i \sqrt{\frac{\gamma_i}{\gamma_i + \sigma_j^2}} \exp\left(-\frac{(\bar{z}_j + \delta - z_i)^2}{2(\gamma_i + \sigma_j^2)}\right) \exp\left(-\frac{(\theta_i - z_i)^2}{2\alpha_i}\right). \tag{6}$$

S1.2: Phenotypic response to selection

Denoting the population mean fitness by $\overline{W}_i$ and the phenotypic distribution of species i by $\phi_i(z_i)$, we have

$$\overline{W}_i = \int_{\mathbb{R}} W_i(z_i)\phi_i(z_i)dz_i. \tag{7}$$

Using equation (6) from Appendix S1.1 we have

$$\overline{W}_i = K_i \exp\left(-\frac{\alpha_i(\bar{z}_j + \delta - \bar{z}_i)^2 + (\gamma_i + \sigma_j^2)(\theta_i - \bar{z}_i)^2}{2(\gamma_i\alpha_i + \gamma_i\sigma_j^2 + \gamma_i\sigma_i^2 + \alpha_i\sigma_j^2 + \alpha_i\sigma_i^2 + \sigma_i^2\sigma_j^2)}\right), \tag{8}$$

where K_i is independent of both $\bar{z}_i$ and $\bar{z}_j$. Once again, we have capitalized on two iterations of (4) to obtain (8). We can now employ our weak selection approximations by assuming both selective tolerances α_i and γ_i are sufficiently large so that $\gamma_i + \sigma_j^2 + \sigma_i^2 \approx \gamma_i$ and $\alpha_i + \sigma_j^2 + \sigma_i^2 \approx \alpha_i$. Mean fitness is then approximated as

$$\overline{W}_i \approx K_i \exp\left(-\frac{(\bar{z}_j + \delta - \bar{z}_i)^2}{2\gamma_i} - \frac{(\theta_i - \bar{z}_i)^2}{2\alpha_i}\right). \tag{9}$$

Hence, the breeders equation (sensu Lande, 1976) returns

$$\Delta\bar{z}_i = G_i \frac{\partial \ln \overline{W}_i}{\partial \bar{z}_i} \approx G_i\left(\gamma_i^{-1}(\bar{z}_j + \delta - \bar{z}_i) + \alpha^{-1}(\theta_i - \bar{z}_i)\right) \tag{10}$$

where G_i is the additive genetic variance for the trait in species i. For simplicity, we assume that the G_i are constant with respect to time for both species. We define the strength of abiotic selection as $A_i \equiv \alpha_i^{-1}$. Similarly we define the strength of biotic selection by $B_i \equiv \gamma_i^{-1}$. Hence, the response of the population mean phenotype to selection and reproduction is approximated by

$$\Delta\bar{z}_i \approx G_i\left(B_i\delta + B_i(\bar{z}_j - \bar{z}_i) + A_i(\theta_i - \bar{z}_i)\right). \tag{11}$$

S1.4: Relating selection strengths to selection gradients

A selection gradient, commonly denoted by β, provides an alternative measure of selection. It occurs in the breeders equation given by $\Delta\bar{z} = G\beta$ (Lande and Arnold, 1983). Assuming that independent sources of selection can be decomposed (Ridenhour, 2005) so that $\beta = \beta_B + \beta_A$ represents the decomposition of selection into "biotic" and "abiotic" sources then, in terms of the offset matching model, we have

$$\beta_{B,i} = B_i\delta + B_i(\bar{z}_j - \bar{z}_i),\ \beta_{A,i} = A_i(\theta_i - \bar{z}_i). \tag{12}$$

S1.5: Drift

Drift is modeled as the response to randomly sampling the offspring population. Denoting $Z_1, \ldots, Z_{n_i}$ as the breeding values of a random sample of individuals drawn from species i where n_i is the effective population size, we can calculate the random variable $\tilde{z}_i = \frac{1}{n_i}\sum_{k=1}^{n_i} Z_k$. Assuming these breeding values are drawn independently of one another and that their distribution is normal with mean $\bar{z}_i$ and variance $h_i^2\sigma_i^2 = G_i$ (h_i^2 denotes heritability), we have $\tilde{z}_i$ is distributed normally with mean $\bar{z}_i$ and variance G_i/n_i. We denote the response to drift in species i by $\xi_i \equiv \tilde{z}_i - \bar{z}_i$.

In the case that n_i is different across N populations we must alter our treatment of drift. We denote by n_{ij} the effective population size of species i in population j and Z_{ijk} the trait of individual k in this population. At first it may seem that replacing ξ_i with the average $\frac{1}{N}\sum_j \xi_{ij}$ would be appropriate. However this average reduces in variance as N grows. The effect of drift on local populations should not attenuate as sample size increases. We therefore instead take motivation from the Lindeberg-Lévy central limit theorem and define $\tilde{\xi}_i \equiv \frac{1}{\sqrt{N}}\sum_j \xi_{ij}$ as our "effective" response to drift. Under this definition we have

$$\mathrm{Var}(\tilde{\xi}_i) = \frac{1}{N}\sum_j\sum_k \mathrm{Var}(Z_{ikj}) = \frac{1}{N}\sum_j \frac{G_i}{n_{ij}} = G_i\frac{\sum_j n_{ij}^{-1}}{N}. \tag{13}$$

Hence, if there are multiple estimates of effective population size available from different locations (such as for our camellia-weevil example) we recommend using the harmonic mean of these values. Likewise, to calculate the effective additive genetic variance for our model when the additive genetic variance is known to be different across space we repeat the above approach to obtain

$$\mathrm{Var}(\tilde{\xi}_i) = \frac{1}{N}\sum_j\sum_k \mathrm{Var}(Z_{ikj}) = \frac{1}{N}\sum_j \frac{G_{ij}}{n_i} = \frac{1}{n_i}\frac{\sum_j G_{ij}}{N}. \tag{14}$$

Thus, we recommend using the arithmetic average of additive genetic variances when they vary across space.

Beyond the change in mean phenotype, we ignore the effects of drift on variance or any other property of the population. Then the total change in the mean trait of the population for species i after selection, reproduction and drift is

$$\Delta\bar{z}_i = G_i\left(B_i\delta + B_i(\bar{z}_j - \bar{z}_i) + A_i(\theta_i - \bar{z}_i)\right) + \xi_i. \tag{15}$$

S1.6: Metapopulation dynamics

We denote by the random vector $(\bar{Z}_1(t), \bar{Z}_2(t))$ the pair of trait means for each species within a given (but unspecified) population at time step t. Motivated by the result for local mean

trait dynamics, we define the next generation mean trait pair by

$$\bar{Z}_i(t+1) \equiv \bar{Z}_i(t) + G_i \left[B_i\delta + B_i(\bar{Z}_j(t) - \bar{Z}_i(t)) + A_i(\theta_i - \bar{Z}_i(t)) \right] + \xi_i(t). \tag{16}$$

This defines a two dimensional random process. We describe the statistical behavior of this process by focusing on its distribution. In general, to fully describe the distribution of a random process we need to write down all of the statistical moments at each time step. Of course this is not feasible, but, as shown in Appendix S1.8, for our model we only need to calculate the first five moments. The first two are the average values of the two local mean traits (μ_1, μ_2) across the entire metapopulation. The third and fourth moments are the variances of the local mean traits among the metapopulation (V_1, V_2), which can take on positive values in our model solely due to drift. Finally, the fifth moment is the spatial covariance of the two mean traits (C) which can be non-zero in our model due to either $B_1 \neq 0$, $B_2 \neq 0$ or both $B_1, B_2 \neq 0$. Then to construct the metapopulation model we identify $\mu_i(t) \equiv \mathbb{E}\bar{Z}_i(t)$, $V_i(t) \equiv \mathbb{E}[(\mu_i(t) - \bar{Z}_i(t))^2]$, and $C(t) \equiv \mathbb{E}[(\mu_1(t) - \bar{Z}_1(t))(\mu_2(t) - \bar{Z}_2(t))]$. The simplicity of the local dynamics makes it a straightforward calculation to show that this definition coincides with the recursions

$$\mu_i' = \mu_i + G_i \left[B_i\delta + B_i(\mu_j - \mu_i) + A_i(\theta_i - \mu_i) \right] \tag{17a}$$

$$V_i' = (1 - A_iG_i)^2 V_i + 2(1 - A_iG_i)B_jG_j(C - V_i) + B_i^2G_i^2(V_i + V_j - 2C) + G_i/n_i \tag{17b}$$

$$\begin{aligned} C' = {} & (1 - A_1G_1)(1 - A_2G_2)C + (1 - A_1G_1)B_2G_2(V_1 - C) + (1 - A_2G_2)B_1G_1(V_2 - C) \\ & + B_1B_2G_1G_2(2C - V_1 - V_2). \end{aligned} \tag{17c}$$

Taking this approach makes the implicit assumption of infinitely many localities. This is due to the fact that the moments are calculated are over all possible sample paths. Each sample path is associated with some theoretical location. However, in the real world metapopulations are comprised of just a finite number of localities. Then, in view of our model, they provide just a subset of all the possibilities. The true metapopulation moments of those empirical systems would then be looked at as sample moments with respect to the theoretical metapopulation moments predicted by our model. These observations dove-tail nicely with the fact that true empirical metapopulation moments exhibit some stochastic behavior while the metapopulation moments of our model evolve deterministically.

Finally, we implement simplifications of the expressions (17) by assuming each B_i and A_i are of some small order $\epsilon \ll 1$ and neglect all terms of order ϵ^2 and higher.

S1.8: Proof of normality

Here we demonstrate that as long as selection is weak, the distribution describing population mean phenotypes across the metapopulation converges to a bivariate normal distribution regardless of the initial distribution of the metapopulation. Our model is 2-dimensional, but the proof is easily obtained for the more general d-dimensional situation. Using $\|x\| = \sum_{i=1}^{d} |x_i|$ for $x \in \mathbb{R}^d$ and $\|U\| = \max\{|\lambda| : \lambda \text{ is an eigenvalue of } U\}$ for matrices, we have the following:

Claim: If $X(t)$ is a d-dimensional random process defined by $X(t+1) = UX(t) + W(t)$ where U is a $d \times d$ nonsingular matrix with real entries, $\|U\| < 1$ and the sequence $W(t)$ is comprised of iid d-dimensional multivariate normal variables, then $X(t)$ will converge in distribution to a multivariate normal random variable regardless of the distribution of $X(0)$.

Proof: For clarity we move the time index t to the subscript. The solution is immediately obtained as $X_t = U^t X_0 + \sum_{k=0}^{t-1} U^k W_k$. The hypothesis implies $\lim_{t\to\infty} U^t = 0$, the $d \times d$ zero matrix. So, we have $X^* \equiv \lim_{t\to\infty} X_t = \sum_{k=0}^{\infty} U^k W_k$. Note that $\|U^k\| \leq \|U\|^k$ for each $k = 0, 1, \ldots,$ that $\mathbb{E}\|W_i\| = \mathbb{E}\|W_j\| < \infty$ for each $i, j = 0, 1, \ldots,$ and that $\sum_{k=0}^{\infty} \|U\|^k = (1 - \|U\|)^{-1} < \infty$. Hence, using Tonelli's Theorem to reverse the order of the expectation and summation, we have

$$\mathbb{E}\|X^*\| = \mathbb{E}\Big\| \sum_{k=0}^{\infty} U^k W_k \Big\| \leq \mathbb{E}\left(\sum_{k=0}^{\infty} \|U^k\| \|W_k\| \right)$$

$$= \mathbb{E}\|W_0\| \sum_{k=0}^{\infty} \|U^k\| \leq \mathbb{E}\|W_0\| \sum_{k=0}^{\infty} \|U\|^k < \infty.$$

This demonstrates that the mean vector of X^* has finite magnitude. Denote by $\mathrm{Var}(X)$ the variance covariance matrix of a d-dimensional random vector X. Then, by the iid assumption on the W_t we have

$$\|\mathrm{Var}(X^*)\| = \Big\| \sum_{k=0}^{\infty} \mathrm{Var}(U^k W_k) \Big\| = \Big\| \sum_{k=0}^{\infty} U^k \mathrm{Var}(W_0) (U^k)^\top \Big\| \leq \sum_{k=0}^{\infty} \|U^k\| \|\mathrm{Var}(W_0)\| \|U^k\| = \star.$$

By hypothesis $\|U\| < 1$ which implies $\|U^k\|^2 \leq \|U\|^k$ for each $k = 1, 2, \ldots$. Hence,

$$\star = \|\mathrm{Var}(W_0)\| \sum_{k=0}^{\infty} \|U^k\|^2 \leq \|\mathrm{Var}(W_0)\| \sum_{k=0}^{\infty} \|U\|^k < \infty.$$

This shows that $\mathrm{Var}(X^*)$ has finite entries. The higher cumulants of X^* will be zero since the W_t contribute nothing more than second order cumulants, which is characteristic of the

multivariate normal distribution (Bryc, 1995; Lukacs, 1970). Hence, X^* is a d-dimensional multivariate normal random variable. $\square$

In the notation used in the above proof, our model of coevolution corresponds to

$$U = \begin{pmatrix} 1 - G_1(B_1 + A_1) & G_1 B_1 \\ G_2 B_2 & 1 - G_2(B_2 + A_2) \end{pmatrix}. \tag{18}$$

The eigenvalues λ of the above matrix satisfy

$$\begin{aligned} |\lambda| \leq \frac{1}{2}|2 - G_1(A_1 + B_1) - G_2(A_2 + B_2)| + \\ \frac{1}{2}\sqrt{4G_1G_2B_1B_2 + [G_1(A_1 + B_1) - G_2(A_2 + B_2)]^2} \end{aligned} \tag{19}$$

By construction of our model we assume $A_i > 0$. If we futher impose $0 < G_i(B_i + A_i) < 1$ it can be demonstrated with basic calculus that the right hand side of the above inequality is restricted to the half-open unit interval $[0, 1)$. Hence, these conditions are sufficient for our model to satisfy the above claim. This allows us to focus on just the first and second order moments without losing any information predicted by the model.

Appendix S2: Maximum likelihood

In this section we describe our use of maximum likelihood for estimating model parameters defining the strength of coevolution.

S2.1: Estimating selection parameters and hypothesis testing

To connect our model to data, we assume that the metapopulation dynamics have reached equilibrium in distribution. This means that the distribution of the random variables representing the mean traits of each species within a given population is not changing. This assumption allows us to write the means and variances as static functions of the model parameters. By assuming the equilibrium distribution of the metapopulation is bivariate normal, the likelihood of the data (the mean traits of each species in each population; $D = \{D_i\}_{i=1}^N \equiv \{(\bar{z}_1, \bar{z}_2)_i^\top\}_{i=1}^N$) given the two metapopulation mean trait values ($\mu = (\mu_1, \mu_2)^\top \equiv \mathbb{E}(\bar{Z}_1, \bar{Z}_2)^\top$), the two metapopulation trait variances ($V_1 \equiv \mathrm{Var}(\bar{Z}_1),\ V_2 \equiv \mathrm{Var}(\bar{Z}_2)$), and the metapopulation spatial covariance of the two traits ($C \equiv \mathrm{Cov}(\bar{Z}_1, \bar{Z}_2)$), ie, the first five moments, can be expressed as

$$L(D|\mu, \Sigma) = \prod_{i=1}^{N} \frac{\exp\left(-\frac{1}{2}(D_i - \mu)^\top \Sigma^{-1}(D_i - \mu)\right)}{2\pi\sqrt{|\Sigma|}},\ \Sigma = \begin{pmatrix} V_1 & C \\ C & V_2 \end{pmatrix}. \tag{20}$$

Using equations (17) we can easily derive the following equilibrium solutions of the first five moments

$$\mu_i = \frac{A_i A_j \theta_i + A_i B_j \theta_i + A_j B_i (\theta_j + \delta) + 2 B_i B_j \delta}{A_i A_j + A_i B_j + A_j B_i} \tag{21a}$$

$$V_i = \frac{B_i C + \frac{1}{2n_i}}{A_i + B_i} \tag{21b}$$

$$C = \frac{B_1(A_1 + B_1)G_1 n_1 + B_2(A_2 + B_2)G_2 n_2}{2(A_1 A_2 + A_1 B_2 + A_2 B_1)((A_1 + B_1)G_1 + (A_2 + B_2)G_2)n_1 n_2}. \tag{21c}$$

These expressions can in turn be used to solve for the two strengths of abiotic selection, the two strengths of biotic selection and the optimal offset. Because these expressions are quite cumbersome, however, we refer the reader to the Mathematica notebook associated with this paper for their full expressions. However, for the sake of developing intuition and for later use, we show the expressions of just the selection strengths as functions of other model parameters and statistical moments

$$A_i = \frac{\mu_j + \delta - \mu_i}{2n_i(V_i(\mu_j + \delta - \theta_i) - C(\mu_i - \theta_i))} \tag{22a}$$

$$B_i = \frac{\mu_i - \theta_i}{2n_i(V_i(\mu_j + \delta - \theta_i) - C(\mu_i - \theta_i))}. \tag{22b}$$

Since the sample moments of a multivariate normal distribution are the maximum likelihood estimates of its underlying moments and our solutions for the selection strengths and optimal offset correspond precisely with the sample moments, our solutions also maximize likelihood of our model given the data. That is, they maximize the function

$$L(D|A_1, A_2, B_1, B_2) \equiv L(D|\mu(A_1, A_2, B_1, B_2), \Sigma(A_1, A_2, B_1, B_2)). \tag{23}$$

Restricting the selection parameters B_i to match our null hypotheses of unilateral evolution ($B_1 = 0$ or $B_2 = 0$) results in a restricted likelihood (specifically, $L(D|A_1, A_2, 0, B_2)$ or $L(D|A_1, A_2, B_1, 0)$), one whose value is always less than that of $L(D|A_1, A_2, B_1, B_2)$. Then to calculate the significance of B_1, for example, we compute the probability that the statistic

$$\Lambda_1 = 2 \ln \left(\frac{L(D|A_1, A_2, B_1, B_2)}{L(D|A_1, A_2, 0, B_2)} \right) \tag{24}$$

lands within a given percentile of its distribution under the assumption of the null hypothesis $B_1 = 0$. We adopt the standard 95th percentile as our threshold for determining significance (hence, $\alpha = 0.05$). According to Wilk's Theorem, the distribution of this statistic is approximated by a Chi-square (Wilks, 1938). We therefore use this distribution to approximate our p-values.

We now use the results of this section to derive expressions for the restricted maximum likelihood moments under the two null conditions just described. Note that according to equation 22b, when $B_i = 0$ we must have $\mu_i = \theta_i$ for otherwise some parameter or moment must be infinite. This leaves one parameter fixed and one statistical moment fixed. The remaining four parameters can be solved for in terms of the remaining four statistical moments. Without loss of generality, we consider the situation when $B_1 = 0$. This results in the following solutions for the remaining strengths of selection and optimal offset

$$A_1 = \frac{1}{2n_1V_1} \tag{25a}$$

$$A_2 = \frac{1}{C^2 - V_1V_2}\left(\frac{C-V}{2n_2} + \frac{CG_1(V_2 - C)}{2n_1G_2V_1}\right) \tag{25b}$$

$$B_2 = \frac{1}{C^2 - V_1V_2}\left(-\frac{C}{2n_2} - \frac{CG_1V_2}{2n_1G_2V_1}\right) \tag{25c}$$

$$\delta = (\theta_2 - \theta_1) + (\mu_2 - \theta_1)\frac{G_2n_1V_1^2 + G_1n_2C^2}{G_2n_1V_1C + G_1n_2V_2C} \tag{25d}$$

That the remaining four equations for the equilibrium of statistical moments are invertible with respect to the remaining selection strengths and optimal offset indicates that the only moment whose value changes under the restricted maximum likelihood when $B_1 = 0$ is μ_1. Likewise, when we fix $B_2 = 0$ the only change in the resulting maximum likelihood moments is $\mu_2 = \theta_2$. Hence, formulating the likelihood ratios used to estimate the significance of the B_i are relatively straightforward.

Appendix S3: Inference under broken assumptions

In this appendix we evaluate the performance of our method when some key assumptions are broken. Specifically, we extend the analyses described in the main text to inference when gene-flow is present, when data are significantly non-normal and when there is error in estimating the abiotic optima.

S3.1: Inference for systems with gene-flow

Although our method formally assumes the absence of gene-flow among populations, we anticipate its usefulness persists in the face of weak gene-flow. To determine the validity of this assertion we extend our model to include gene-flow as captured by an island model. Under this model the effect of gene-flow is to reduce the metapopulation variances (and hence covariance, but not correlation). As both strengths of selection also act to reduce

variance across the metapopulation, we expect the inclusion of gene-flow to negatively bias our estimates of biotic selection.

In terms of the local model, the inclusion of gene-flow is approximated by an additional linear term. Denoting the strength of gene-flow by m_i, we have

$$\Delta\bar{z}_i = G_i(A_i(\theta_i - \bar{z}_i) + B_i\delta + B_i(\bar{z}_j - \bar{z}_i)) + m_i(\mu_i - \bar{z}_i) + \xi_i. \tag{26}$$

Assuming weak migration along with weak selection, we obtain

$$\Delta\mu_i = G_i\left[B_i\delta + B_i(\mu_j - \mu_i) + A_i(\theta_i - \mu_i)\right] \tag{27a}$$

$$\Delta V_i = -2(A_iG_i + m_i)V_i + 2B_jG_j(C - V_i) + G_i/n_i \tag{27b}$$

$$\Delta C = B_2G_2(V_1 - C) + B_1G_1(V_2 - C) - (A_1G_1 + A_2G_2 + m_1 + m_2)C. \tag{27c}$$

We can then solve for the selection strengths at equilibrium to arrive at

$$A_i = \left(\frac{1}{2n_i} - \frac{m_iV_i}{G_i}\right)\left(\frac{\mu_j + \delta - \mu_i}{V_i(\mu_j + \delta - \theta_i) - C(\mu_i - \theta_i)}\right) \tag{28a}$$

$$B_i = \left(\frac{1}{2n_i} - \frac{m_iV_i}{G_i}\right)\left(\frac{\mu_i - \theta_i}{V_i(\mu_j + \delta - \theta_i) - C(\mu_i - \theta_i)}\right). \tag{28b}$$

From these expressions we see that if the additive genetic variance within populations is much greater than the variance in mean phenotype among populations, gene-flow can safely be ignored. Of course, this will not likely be the case. Hence, we repeat our analysis of method performance described in the main text for data simulated with gene-flow present. In Figure 1 of this appendix we display error rates for detecting coevolution and regression statistics for inferring the quantity of coevolution as functions of the amount of gene-flow present. These results demonstrate that low rates of gene-flow have negligible effects on our methods ability to infer coevolution, but that more extreme rates of gene-flow (e.g. 0.01) significantly reduces our methods precision in inferring the strength of coevolution as demonstrated by a reduced R^2.

S3.2: Inference using non-normal data

The requirement that phenotypic data be bivariate normal will likely restrict the applicability of this method. However, if this method can be shown to be robust to non-normal data, the range of potentially coevolving systems for which this method can be used will expand greatly. We are therefore motivated to investigate the effect of non-normal data on inference. To understand this effect we repeat the methods discussed in the main text for simulated non-normal data sets.

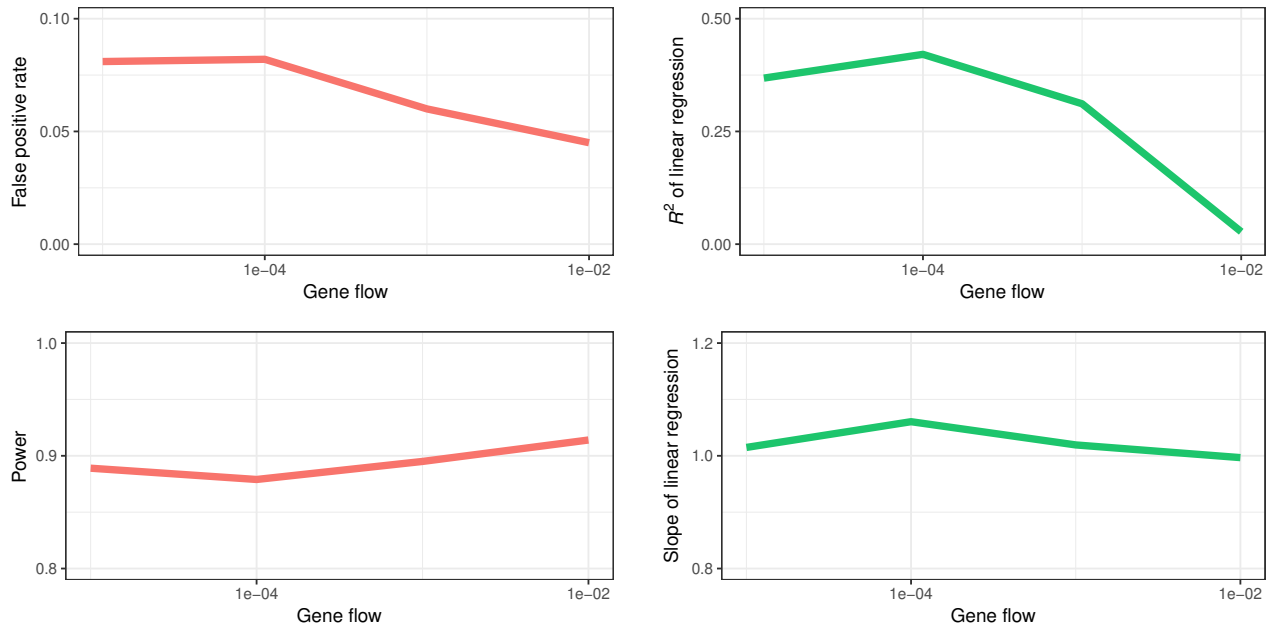

Figure 1: Error rates and regression statistics as functions of the strength of gene-flow.

To achieve this we repeatedly simulated data under our model until the resulting data set failed to be normal. We used the Shapiro-Wilks test with a threshold of normality τ to determine whether or not the data are normal (R Core Team, 2016). That is, if the p-value returned by the Shapiro-Wilks test is less than τ, we conclude the data to be non-normal. Using only those data sets which were determined to be significantly non-normal, we proceeded to perform inference as described above. We performed regression analysis on our estimates of biotic selection strengths and analysed false positive rates and power across a range of threshold values. Results are displayed in Figure 2 of this appendix. From this figure we see the effect of non-normality on inference is negligible, even for highly non-normal data.

S3.3: Inference with measurement error

We analyzed the effect of measurement error in the background parameters has on our ability to infer coevolution. In doing so we focused on error in measuring abiotic optima. We assume the estimates of abiotic optima are normally distributed and centered on the "true" optima. Using the same approach for performance analysis as described in the main text, we examine regression statistics and error rates as functions of the standard deviation in the estimated abiotic optima. Results are displayed in Figure 3 of this appendix. These results demonstrate that our method of coevolutionary inference is insensitive to measurement error in the abiotic

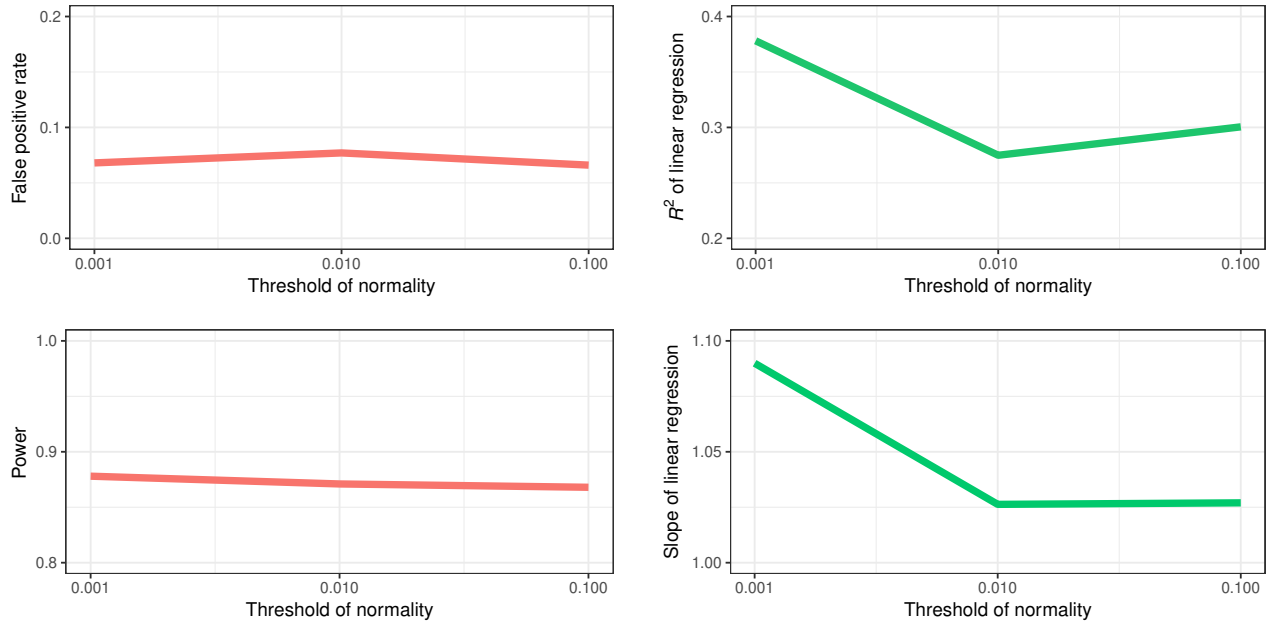

Figure 2: Error rates and regression statistics as functions of τ, the threshold for which data is determined to be non-normal. The smaller τ is the more non-normal the data is according to the Shapiro-Wilks test.

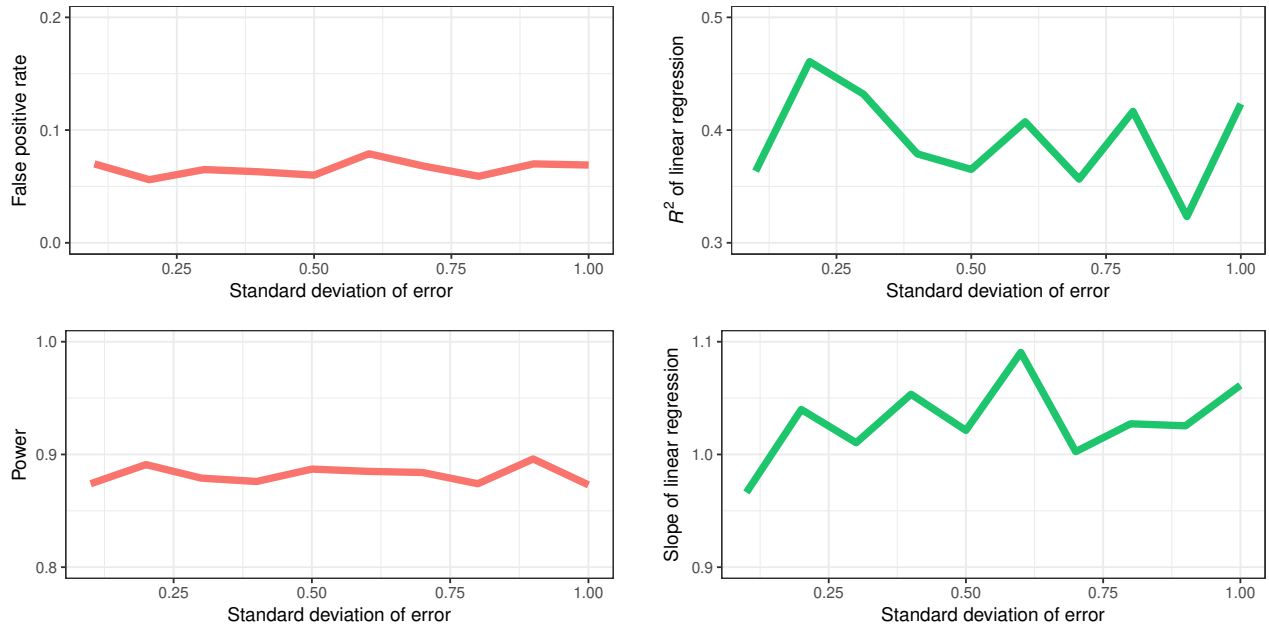

Figure 3: Error rates and regression statistics as functions of the amount of measurement error in abiotic optima.

optima.

Appendix S4: Analysis of data

Here we describe the data and methods used for the analysis of coevolution. At its most basic level, our approach requires estimates of the mean phenotypes for both species in at least two populations where they are known to interact. This data is used to calculate the metapopulation means, variances and covariance upon which our maximum likelihood estimation rests. In addition to this core data, our approach requires estimates for several key background parameters: the effective local population sizes for each species n_i, the optimal traits in the absence of the interaction θ_i (the abiotic optima) and the additive genetic variances G_i. Model parameters that are inferred by our method are the strengths of abiotic selection A_i, the strengths of biotic selection B_i, and the optimal offset δ.

For many systems, it will be possible to estimate effective population sizes using molecular techniques (e.g. Beerli, 2005). If these are known for each population, or for a subset of populations, the harmonic mean of these should be used for the parameter n_i. This is justified in Appendix S1. In the absence of effective population size estimates, one can instead infer the composite parameters $A_i n_i$ and $B_i n_i$. As shown in equation 22b the selection strengths are directly proportional to the effective population sizes. Hence, the suggested composite parameters provide reasonable proxies for the strengths of selection.

The abiotic optima are particularly important for inference. If these parameters are close to the biotic optima, it becomes difficult to assess the relative importance of biotic selection as its signal becomes partially masked by the patterns resulting from abiotic selection. In our analyses, three distinct approaches were used to estimate these parameters: (1) We used the trait means of populations isolated from the interacting species as an estimate for θ_i. However, if there is significant gene flow between these populations and populations where the interaction is taking place, the resulting estimate for θ_i may be biased towards the biotic optimum, which would weaken the coevolutionary signal and make our approach conservative. Also, this approach assumes the form and strength of abiotic selection is equal in locations where the interaction does and does not occur. (2) We also used the mean phenotypes of members of a species that do not partake in the interaction at any point in their life history to estimate θ_i. For example, in the camellia-weevil interaction, the males do not oviposit and hence do not interact with camellia by boring holes in its pericarp at any point in their life histories. Hence, we assume their rostrum lengths provide decent estimates of the abiotic optima. This approach also assumes approximately equivalent abiotic selection surfaces for both individuals within a population that do and do not partake in the interaction. (3) Lastly, we estimated the abiotic optimum for the fly *M. longirostris* through data on closely related species. We reason that each sibling species represents an alternative

evolutionary trajectory, especially since they do not coincide with the *M. longirostris* pollination guild (Barraclough and Slotow, 2010). Hence, we may crudely estimate the abiotic optimal phenotype as the trait values of these sibling species.

In the following sections, we describe how we applied this general approach to interactions between the flower *Lapeirousia anceps* and its fly pollinator *Moegistorhynchus longirostris* as well as the camellia fruit *Camellia japonica* and its weevil seed-predator *Curculio camelliae*. In both interactions, values subscripted with the number 1 indicate the insect species and values subscripted with the number 2 the plant species. All computations were performed in the statistical programming language *R* (R Core Team, 2016). Data and scripts used in this analysis will be made available at:

`https://github.com/bobweek/measuring.coevolution`

S4.1: The interaction between *L. anceps* and *M. longirostris*

Phenotypic data was obtained from previously published work (Pauw et al., 2009) estimating the population mean proboscis length and population mean tube length for eight populations in South Africa where this interaction occurs. These data are summarized in Table 1.

The abiotic optimum trait value for *L. anceps* was assumed to be equal to the average of the mean tube lengths found in Red Hill and Uitkomsberge, the only two populations recorded which are not visited by *M. longirostris* (Pauw et al., 2009). This average came to be 41.45 mm. For the fly, we estimated the abiotic optimum in four different ways. For the first three we set the abiotic optimum equal to the recorded trait-value of one of the sister species (found in Bequaert, 1935): (1) *M. braunsi*, (2) *M. brevirostris* and (3) *M. perplexus*. Only for *M. braunsi* were separate measurements recorded for each sex. We therefore use the average of these two lengths as the species wide mean. (4) Lastly, we estimated the abiotic optimum as the average of the mean traits for the three sibling species. These estimates are summarized in Table 2. We implemented our method for measuring coevolution separately for each of these four different parameterizations. Table 2 shows the results under each of these parameterizations do not substantially differ.

Table 1: Moments used for analysis in the fly-flower interaction.

Moments	Sample Moments
μ_1	65.83 mm
μ_2	55.74 mm
V_1	206.4 mm^2
V_2	163.6 mm^2
C	143.4 mm^2

Table 2: Results under different abiotic optima for *M. longirostris*.

	M. brevirostris	*M. perplexus*	*M. braunsi*	Average
θ_1	11.5 mm	32.0 mm	41.0 mm	28.17 mm
B_1	6.409660e-05 mm^{-2}	6.394011e-05 mm^{-2}	6.386160e-05 mm^{-2}	6.397159e-05 mm^{-2}
B_2	1.718896e-06 mm^{-2}	1.867071e-06 mm^{-2}	1.941416e-06 mm^{-2}	1.837269e-06 mm^{-2}
A_1	7.025746e-06 mm^{-2}	7.073487e-06 mm^{-2}	7.097440e-06 mm^{-2}	7.063885e-06 mm^{-2}
A_2	3.140721e-06 mm^{-2}	3.122435e-06 mm^{-2}	3.113260e-06 mm^{-2}	3.126113e-06 mm^{-2}
δ	16.03591 mm	13.82320 mm	12.84025 mm	14.23957 mm
p_1	<2.22e-16	<2.22e-16	2.22e-16	<2.22e-16
p_2	1.19e-07	1.19e-07	1.19e-07	1.19e-07
$\mathfrak{C}$	1.049645e-05 mm^{-2}	1.092615e-05 mm^{-2}	1.113472e-05 mm^{-2}	1.084126e-05 mm^{-2}
$\mathfrak{B}$	0.1745277	0.1861598	0.1918956	0.1838421

S4.2: The interaction between *C. camelliae* and *C. japonica*

Phenotypic data was obtained from previously published work (Toju et al., 2011a) estimating the population mean rostrum length and population mean pericarp thickness for weevil and camellia populations throughout Japan. We follow the approach of Toju et al. (2011a) by focusing on female weevils as male weevils do not oviposit and hence are not involved with the interaction. Of the phenotypic data available we included only those populations for which each species was sampled. This made for 25 populations (see the csv for more details). Table 3 summarizes the metapopulation means, variances and covariance of these data.

Effective population sizes were taken as the harmonic means of those reported in Toju et al. (2011b). We assumed the male weevil rostrum length to be the abiotic optimum of the weevil species. Means for the rostrum length of male weevils are available from 11 different populations (Nagaoka, Kyoto, Jurinji, Taiji, Muroto, Ashizuri, Reihoku, Takahama,

Yahazu, Shitoko and Hyanyama). The optimal pericarp thickness for the camellia was estimated as the average of mean pericarp thicknesses from 17 populations with no recorded weevil populations (Miyatsuka, Hatake, Mukai, Izuhara, Kamitsuki, Izu, Igatani, AkoN2, Tsubota, Jinoshima, Kishiku, Tomie, Kuwanoura, Ichinoura, Nakakoshiki, Kashima and Teuchi). These values were taken from Toju et al. (2011a). The resulting background parameters can be found in Table 4.

Table 3: Moments used for analysis in the camellia/weevil interaction.

Moments	Sample Moments
μ_1	12.5 mm
μ_2	10.4 mm
V_1	14.2 mm^2
V_2	28.8 mm^2
C	19.3 mm^2

Table 4: Parameter values used for analysis in the weevil-camellia interaction.

Parameters	*C. camelliae*	*C. japonica*
θ_i	5.91 mm	6.26 mm
n_i	3.08e+04	1.79e+03

S4.3: Effect sizes

To estimate the effect of coevolution on the bivariate distribution of mean-trait-pairs, we used our model to predict the parameters of the metapopulation distribution in the absence of coevolution. Referring to the equlibrium expressions 21 in Appendix S2, we can calculate the maximum likelihood solutions for A_1 and A_2 when $B_1 = B_2 = 0$. Under this condition we find $\mu_i = \theta_i$, $C = 0$ and

$$V_i = \frac{1}{2n_i A_i}.$$

Thus, in the absence of coevolution both V_1 and V_2 remain equal to their observed values. The remaining three moments, however, must shift from their observed values.

To calculate the magnitude by which these moments shift, we calculate a metric for the effect size. In general, the effect size of a treatment is typically defined as the difference between sample means with and without the treatment divided by the pooled standard deviation of the samples. A natural multivariate extension of such a metric is the Mahalonobis

distance. Setting μ_c, Σ_c and μ_0, Σ_0 equal to the observed and predicted moments with and without coevolution respectively, we use the following expression as our effect size

$$\mathcal{E} \equiv \sqrt{(\mu_c - \mu_0)^\top \Sigma^{-1} (\mu_c - \mu_0)}$$

where Σ is the pooled covariance matrix, which in our case simplifies to

$$\Sigma = \frac{\Sigma_c + \Sigma_0}{2}.$$

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